Theorem 5.14.3. *Let $N \geq p$ (where p is the number of parts of λ). Then $\chi_{U_\lambda}(C_{\mathbf{i}})$ is the coefficient*[1] *of $x^\lambda := \prod_j x_j^{\lambda_j}$ in the polynomial*

$$\prod_{m\geq 1} H_m(x)^{i_m}.$$

Proof. The proof is obtained easily from the Frobenius formula. Namely, $\chi_{U_\lambda}(C_{\mathbf{i}})$ is the number of elements $x \in S_n$ such that $xgx^{-1} \in P_\lambda$ (for a representative $g \in C_{\mathbf{i}}$), divided by $|P_\lambda|$. The order of P_λ is $\prod_i \lambda_i!$, and the number of elements x such that $xgx^{-1} \in P_\lambda$ is the number of elements in P_λ conjugate to g (i.e., $|C_{\mathbf{i}} \cap P_\lambda|$) times the order of the centralizer Z_g of g (which is $n!/|C_{\mathbf{i}}|$). Thus,

$$\chi_{U_\lambda}(C_{\mathbf{i}}) = \frac{|Z_g|}{\prod_j \lambda_j!}|C_{\mathbf{i}} \cap P_\lambda|.$$

Now, it is easy to see that the centralizer Z_g of g is isomorphic to

$$\prod_m S_{i_m} \ltimes (\mathbb{Z}/m\mathbb{Z})^{i_m},$$

so

$$|Z_g| = \prod_m m^{i_m} i_m!,$$

and we get

$$\chi_{U_\lambda}(C_{\mathbf{i}}) = \frac{\prod_m m^{i_m} i_m!}{\prod_j \lambda_j!}|C_{\mathbf{i}} \cap P_\lambda|.$$

Now, since $P_\lambda = \prod_j S_{\lambda_j}$, we have

$$|C_{\mathbf{i}} \cap P_\lambda| = \sum_r \prod_{j\geq 1} \frac{\lambda_j!}{\prod_{m\geq 1} m^{r_{jm}} r_{jm}!},$$

where $r = (r_{jm})$ runs over all collections of nonnegative integers such that

$$\sum_m m r_{jm} = \lambda_j, \qquad \sum_j r_{jm} = i_m.$$

Indeed, an element of $C_{\mathbf{i}}$ that is in P_λ would define an ordered partition of each λ_j into parts (namely, cycle lengths), with m occurring

[1] If $j > p$, we define λ_j to be zero.